

GENESCAN

DNA SEQUENCE ANALYSIS TOOL USING
BOYER-MOORE STRING MATCHING
ALGORITHM AND NAIVE TEST.



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ABSTRACT

Genescan is a dna sequence analysis system developed using data structures and algorithms to efficiently search for specific dna patterns. The system accepts dna sequences through text input or files such as txt and pdf, extracts the dna sequence, and searches for a selected motif using boyer–moore and naive string matching algorithms. It records important performance measures such as match positions, number of character comparisons, execution time, and shifts. The results of both algorithms are then compared on the same dna sequence to analyze their performance and identify which algorithm performs better for the given input. The project demonstrates how efficient string-matching algorithms can be applied to real-world dna sequence analysis.

PROBLEM STATEMENT

- DNA sequences are very large and often contain specific patterns or motifs that need to be identified efficiently. Traditional searching methods can require many character comparisons, especially when the DNA sequence is long. Therefore, genescan aims to develop a DNA sequence analysis system that uses naive string matching and boyer–moore algorithms to find DNA motifs and compare their performance based on number of comparisons, execution time, match positions, and shifts. The system also supports DNA input from text and files, making sequence analysis easier and more practical.

CORE COMPONENTS



1. DNA Sequence

Input DNA

A, C, G, T nucleotide sequence used for analysis.



2. DNA Motif

Target Pattern

The specific DNA pattern to search within the sequence.



3. String Matching **Boyer—Moore & Naive**

Algorithms used to find the motif and compare their performance.

PROPOSED FUNCTIONALITIES

- **Multiple DNA input methods** – accept DNA sequences through direct text input and files such as TXT, PDF, DOC, and DOCX.
- **Dna sequence extraction & validation** – extract, normalize, and validate dna sequences containing a, c, g, and t.
- **Motif search** – allow users to enter a specific dna motif or pattern to search.
- **Boyer–moore search** – find the motif using bad character and good suffix heuristics.
- **Naive string matching** – perform straightforward pattern matching for comparison.
- **Result visualization** – display match positions, number of matches, comparisons, shifts, and execution time.
- **Algorithm comparison** – compare boyer–moore and naive based on actual performance metrics.
- **Performance insight** – provide a data-based indication of which algorithm performs better for the given dna sequence.
- **Correctness verification** – verify that both algorithms produce the same match positions.

TOOLS & TECHNOLOGIES USED

- **FRONTEND**

- REACT.JS
- HTML5
- CSS3
- JAVASCRIPT
- VITE

- **BACKEND**

- JAVA
- JAVA HTTP SERVER
- REST API

- **ALGORITHMS**

- NAIVE STRING MATCHING
- BOYER–MOORE STRING MATCHING
- BAD CHARACTER HEURISTIC
- GOOD SUFFIX HEURISTIC

- **DEVELOPMENT TOOLS**

- VISUAL STUDIO CODE
- MAVEN
- GIT/GITHUB

OUTPUT

**GeneScan**
DNA SEQUENCE ANALYSIS TOOL

HomeAnalyzeBoyer-MooreComparisonApplications

Start Analysis

BOYER-MOORE STRING MATCHING

Analyze DNA. Find Patterns Faster.

GeneScan is a DNA sequence analysis platform built around the Boyer-Moore string matching algorithm. Search motifs, inspect matches, measure comparisons and visualize the algorithm.

Start DNA Analysis →Learn Boyer-Moore

01DNA INPUT

02MOTIF SEARCH

03DSA TRACE

04PERFORMANCE

ALGORITHM
Boyer-Moore

LIVE SEQUENCE
DNA Pattern Analysis

Ready

01	ATGCGTAC	GTAGCTAG	CTAGCTAG
02	GCTAGCAT	GCTAGCTA	GATCGTAC
03	TAGCTAGC	TAGCTACG	GCTAGCTA
04	CGATCGTA	CGTAGCTA	GATCGATC

PATTERN DETECTED
GCTAGCTA

03
MATCHES

← Previous

▶ Play

Next →

96.5%
FASTER THAN
na Naive Scan

WORKFLOW

- **Step 1 – give DNA input**

“first, the user provides a DNA sequence. We can either paste the DNA directly or upload a TXT, PDF, DOC, or DOCX file.”

- **Step 2 – extract and validate dna**

“the system extracts the dna sequence and checks whether it contains valid dna bases: a, c, g, and t.”

- **Step 3 – enter the motif**

“next, the user enters the dna motif or pattern that they want to find in the sequence.”

- **Step 4 – select the algorithm**

“the user can choose either the boyer–moore algorithm or the naive string matching algorithm.”

- **Step 5 – search the DNA**

“the selected algorithm searches for the motif and finds all the matching positions.”

- **Step 6 – display results**

“the system displays the number of matches, match positions, number of character comparisons, execution time, and shifts for boyer–moore.”

- **Step 7 – run the other algorithm**

“after the first algorithm completes, the user can run the second algorithm on the same dna sequence and motif.”

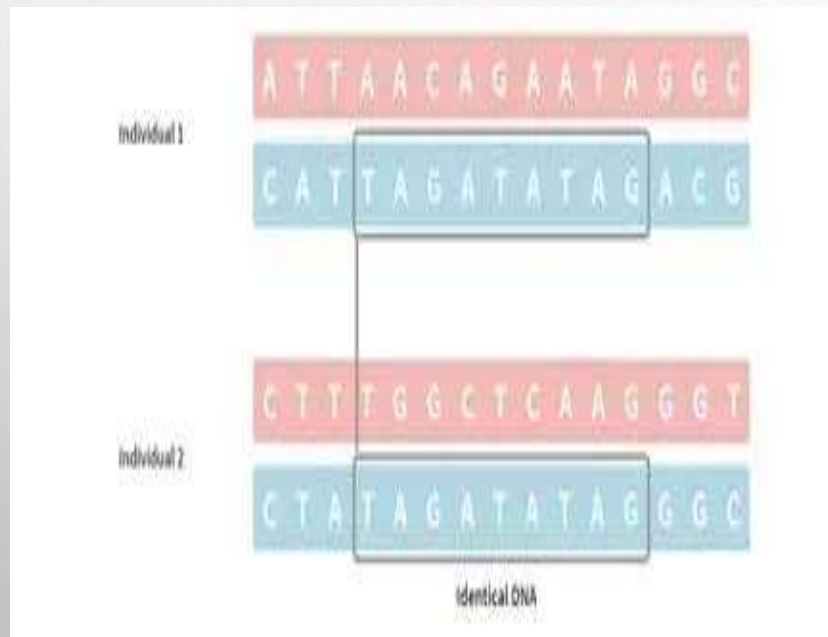
- **Step 8 – compare both algorithms**

“finally, genescan compares both algorithms based on comparisons, execution time, shifts, and correctness.”

- **Step 9 – performance insight**

“based on the actual results, the system gives a performance insight for that particular dna sequence.”

FINAL OUTCOME



- Successfully developed genescan, a DNA sequence analysis system.
- Supports dna input through text and document files.
- Identifies specific dna motifs using boyer–moore and naive string matching.
- Displays match positions, comparisons, execution time, and shifts.
- Provides a direct comparison of both algorithms using the same dna sequence and motif.
- Verifies that both algorithms produce correct and matching results.
- Provides a performance insight based on the measured results for the given dna sequence.
- Demonstrates the practical application of string-matching algorithms in dna sequence analysis.

THANK YOU

